

Smart Search Configuration Guide

Overview

This document covers the configuration settings for **Smart Search**. These settings control how keyword-based, concept (semantic), and image search results are weighted and how the underlying vector similarity engine operates.

Smart Search Weight

The UnifiedSearchWeight parameter controls the relative importance of each search method when blending results. The weight format changes depending on whether image search is enabled. A new index needs to be created if this setting is changed.

Weight Format

Scenario	Format	Parts
With Images	Keyword : Concept : Image	3 parts
Without Images	Keyword : Concept	2 parts

Examples

Configuration	Keyword Weight	Concept Weight	Image Weight
0.3:0.35:0.35	30%	35%	35%
0.5:0.5	50%	50%	N/A

Note: All weight values should sum to **1.0**. Adjust these values based on your use case increase the keyword weight for exact-match-heavy catalogs, or increase concept/image weights for discovery-oriented experiences.

Minimum Score Thresholds

These thresholds filter out low-confidence results from concept and image search. Currently this setting applies only to Concept search and Image search, use the UnifiedSearchWeight parameter to control the blend in Smart search.

Parameter	Default	Description
ConceptSearchMinScore	0	Minimum relevance score a concept (semantic) search result must meet to be included.
ImageSearchMinScore	0	Minimum relevance score an image search result must meet to be included.

Tip: Increasing these values can improve result quality by excluding weak matches but setting them too high may reduce recall.

KNN (K-Nearest Neighbors) Configuration

KNN settings govern the vector similarity search engine used for concept, image and unified search matching. These parameters affect search accuracy, performance, and resource usage.

Knn_Engine

The engine determines which library performs the vector search.

Engine	Status	Description
lucene	 Recommended	The native search library, offering an HNSW implementation with efficient filtering capabilities. Best overall balance of performance and integration.
faiss	Supported	Facebook AI Similarity Search. A comprehensive library implementing both HNSW and IVF methods, with additional vector compression options. Suitable for very large-scale datasets.

Engine	Status	Description
nmslib	⚠️ Deprecated	Legacy engine. Use lucene instead.

Knn_Method

The method defines the algorithm used to organize and search vectors.

Method	Description	Best For
hnsw	Hierarchical Navigable Small World. Creates a hierarchical graph structure of connections between vectors, enabling fast approximate nearest neighbor search. See: Efficient and robust approximate nearest neighbor search using Hierarchical Navigable Small World graphs.	General-purpose use; high recall with low latency.
ivf	Inverted File Index. Organizes vectors into buckets based on clustering. During search, only a subset of buckets is searched, reducing computation.	Very large datasets where some accuracy trade-off is acceptable for speed.

Knn_SpaceType

The space type defines the distance metric used to measure similarity between vectors.

Space Type	Description	Best For
cosinesimil	Cosine Similarity. Measures the angle between two vectors, ignoring magnitude. Two vectors pointing in the same direction are considered similar regardless of length.	NLP and information retrieval where the direction (topic/content) matters more than magnitude (frequency). This is the most commonly used metric for text and semantic search.

Space Type	Description	Best For
l_2	Euclidean Distance (L2 Norm). Measures the straight-line distance between two vector points in space.	Use cases where both direction and magnitude matter, such as image embeddings or spatial data.

Search Tuning Parameters

These parameters fine-tune the balance between search speed and accuracy.

Parameter	Default	Description
ef_search	100	Controls the size of the dynamic candidate list during HNSW search. Higher values improve recall (accuracy) at the cost of increased latency. Lower values speed up search but may miss relevant results.
approximate_threshold	1000	The result count threshold below which approximate search is used. When the number of candidate results exceeds this value, the engine may switch strategies.
filtered_exact_search_threshold	0	Controls when the engine falls back to exact (brute-force) search after filtering. A value of 0 means exact search fallback is effectively disabled. Increase this if filtered queries are returning poor results.

Configuration Summary

Parameter	Default Value	Is new index required for the change to take effect?
UnifiedSearchWeight	0.3:0.35:0.35	Yes
IncludelImageSearchInUnified	0	Yes
ConceptSearchMinScore	0	No
ImageSearchMinScore	0	No
ef_search	100	No
approximate_threshold	1000	No
filtered_exact_search_threshold	0	No
Knn_Engine	lucene	Yes
Knn_SpaceType	cosinesimil	Yes
Knn_Method	hnsw	Yes

Tuning Guidelines

Prioritize Keyword Accuracy

If your catalog relies on exact product names, SKUs, or model numbers, increase the keyword weight:

UnifiedSearchWeight: 0.6:0.2:0.2 (with images)

UnifiedSearchWeight: 0.7:0.3 (without images)

Prioritize Semantic Discovery

If users search with natural language or vague descriptions, increase the concept weight:

UnifiedSearchWeight: 0.2:0.5:0.3 (with images)

UnifiedSearchWeight: 0.3:0.7 (without images)

Prioritize Visual Search

If your product catalog is highly visual (e.g., fashion, home decor), increase the image weight:

UnifiedSearchWeight: 0.2:0.3:0.5 (with images)

Improve Recall at the Cost of Speed

Increase ef_search to 200 or higher for better accuracy in vector search results. Monitor latency impact.